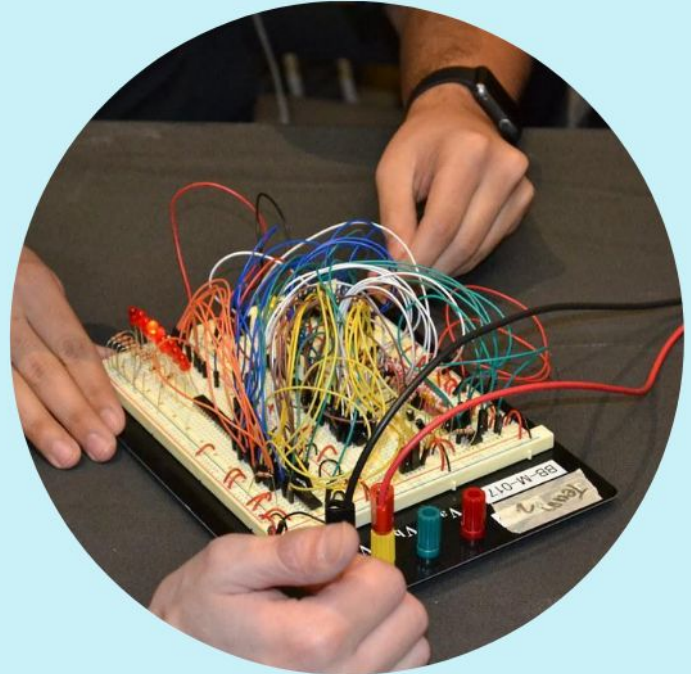


Why Verilog?

What is Electronic Design?

- Process of developing a circuit by using **known** electronic components in order to meet the given **specifications**.
- Specifications: detailed description of desired behavior of the circuit
- Known devices: devices whose behavior can be modeled by known equations or algorithms, with known values of parameters



Hardware Description Languages (HDLs)

- A computer language used to describe the structure and behavior of electronic circuits (usually digital circuits)
- HDL: easy to describe hardware circuits following a particular syntax
- Need for HDLs → Circuits were designed on PCBs, testing and designing of large circuits not easy, slow time to market and complex design flow

→ **Verilog**

→ VHDL (Very High Speed Integrated Circuit Hardware Description Lang.)

Why hardware? Why not more software

In some cases ,like IoT devices ,you may want to run algorithms on systems that don't have a full-fledged computer.

By converting your algorithm into hardware using HDLs, you gain more control over the resources being used, which helps in building efficient, lightweight, and low-power solutions.

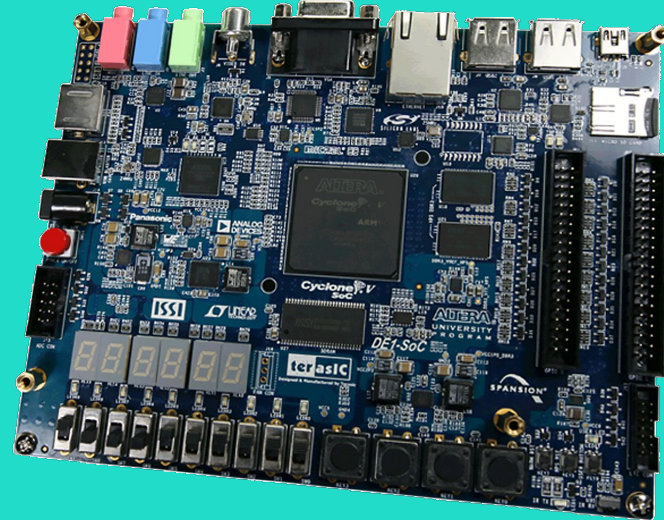
Speed-up with Verilog

Using HDLs like Verilog lets you design hardware that can run parallelly i.e. side by side

So some tasks can be moved from software to hardware, which often makes them run faster and more efficiently, like in areas like signal processing, encryption, or real-time control where speed really matters.

So, this Circuit that we describe using a HDL has to either be printed by a factory or be put in a special kind of programmable hardware called FPGA

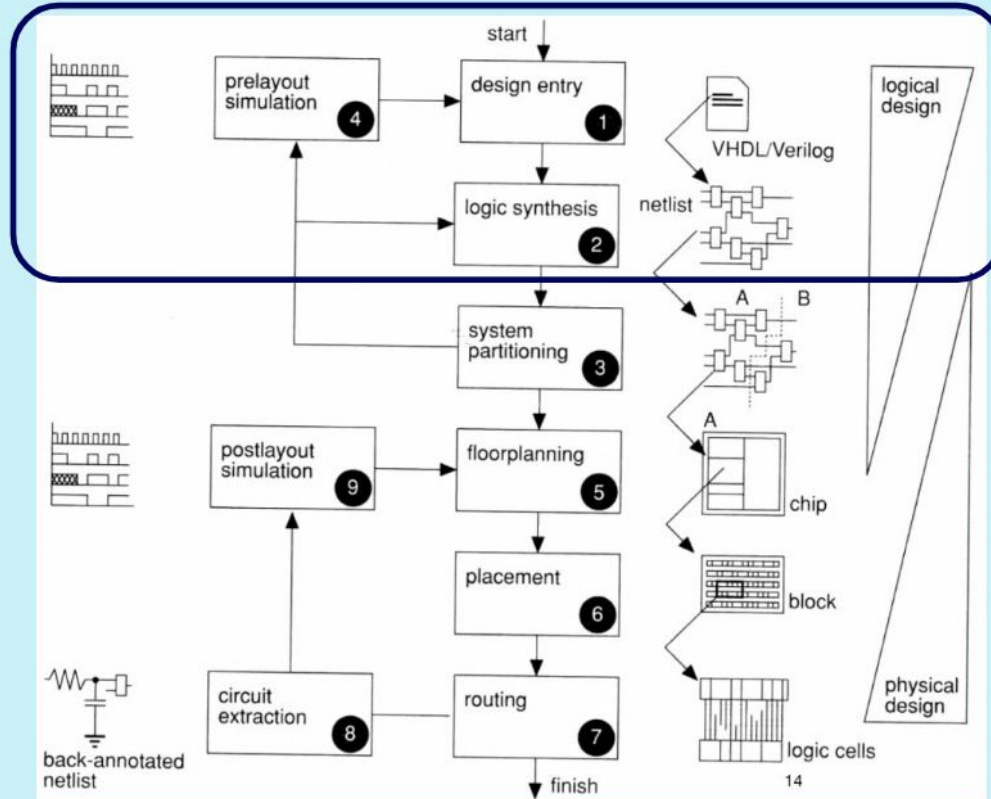
FPGA →



*look at the video linked to see an end to end synthesis

ASIC design flow

What we do as
CS students →



Logic Simulation and Synthesis

- Logic Simulation
 - Runs your circuit in computer before you map it to silicon.
 - Essential for ensuring correctness, testing, etc.
- Logic Synthesis
 - Converts your logic described in high-level to a gate-level description => equivalent to a compiler.
 - Register-transfer level (RTL) to gates conversion
 - Such compilation also involves logic minimization